10. GAMETOGENESIS

DURATION : 02:38

STUDY SHEET

COURSE SUMMARY

This session explores gametogenesis, detailing the crucial differences between the oocyte and the spermatozoon, both cellularly and functionally. Students discover oocyte production, their origin during foetal life, as well as the importance of nurse and follicular cells in embryonic development. The impact of temperature on fertility is also discussed, highlighting the need for warmth for the ovaries and coolness for the testes, as well as the implications for fertility problems in women. Finally, the anatomical links between organs and the arterial system are presented, highlighting the axes of trophic communication and the role of osteopathy in fertility management.

GAMETOGENESIS

COMPARISON OF OOCYTE AND SPERMATOZOON

The **cell volume** of the oocyte is very large, in contrast to that of the spermatozoon which is very small.

We can say that the ovum is in **dilation**, in **opening**, completely in **extension** and open.

Whereas the spermatozoon is in **internal rotation**, **desiccated** and **mummified**.

The **DNA** concentration is very high for spermatozoa.

The **motility** of the oocyte is absent, while the spermatozoon is **active**.

CELL PRODUCTION

The number of oocytes produced is approximately **400**. Primary oocytes are formed during the **foetal period**.

This means that, when you were in your mother's womb, you were already making your own oocytes.

NURSE AND FOLLICULAR CELLS

We distinguish between **nurse cells** and **follicular cells**.

One cell is of the **germline**, the other is of the **somatic line**. One is transmitted through development and the other is given by the mother.

These cells will create **axes**, representing a path for the transmission of **genetic** information, of an **epigenetic** type and of **transgenerational formation**.

TEMPERATURE AND FERTILITY

The **ovaries** need to be inside the body because they require **heat**.

In contrast, the **testicles** have descended outside because they need a cooler **temperature**.

IMPACT ON FEMALE FERTILITY

This is an important point to consider when women consult for **fertility** problems.

Sometimes, when working at the level of the **sacrum**, one can feel a **coldness**, which may indicate a problem.

ANATOMICAL AND PHYSIOLOGICAL LINKS

We will discuss this in relation to the **duodenum**, the **arteries** and the "plicas" or **folds** at the level of the duodenum.

These structures allow the passage of **trophic** information via the arterial system, particularly the **ovarian** or **testicular arteries**.

Fertility often needs to be treated higher up, on the framework of the **facial pentagon** of the **pancreas**.